

Flaxseed Oil Alleviates PFOS-Induced Liver Injury by Regulating Hepatic Cholesterol Metabolism

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ABSTRACT: Perfluorooctanesulfonate (PFOS) is a widespread, persistent environmental pollutant that exerts apparent liver toxicity. Flaxseed oil (FO), a dietary oil rich in α -linolenic acid, has been demonstrated to possess a diverse array of health benefits. However, whether FO protects against PFOS-induced liver injury and its underlying mechanisms remain unclear. C57/BL6 mice were orally treated with different concentrations of FO alone or in combination with 10 mg/kg of PFOS for 28 consecutive days. Blood and liver tissues were collected for proteomic, histopathological, biochemical, immunohistochemical, and molecular examinations. Results demonstrated that FO supplementation reduced PFOS-induced liver injury, as evidenced by a decrease in histopathological changes, serum transaminase (ALT and AST) levels, levels of oxidative stress, and inflammatory cytokine (TNF- α , IL-1 β , and IL-6) levels. Proteomic analyses showed that differentially expressed proteins were enriched in cholesterol metabolic pathways when comparing the PFOS group to the FO supplementation groups. The expression of cholesterol metabolism-related proteins was also subsequently measured, revealing that FO supplementation decreased the protein expressions of SREBP2, HMGCR, and LDLR while increasing the expression of CYP7A1. This study demonstrates that FO can alleviate PFOS-induced hepatotoxicity by regulating hepatic cholesterol metabolism, indicating that FO may serve as an effective dietary intervention for preventing liver injury caused by PFOS.

KEYWORDS: flaxseed oil, PFOS, liver, proteomics, cholesterol metabolism

INTRODUCTION

The role of environmental pollutants as a major source of risk to public health has gradually attracted increasing attention. Perfluorooctanesulfonate (PFOS) is a global pollutant and an environmental toxicant that is ubiquitously distributed in our surroundings.¹ It possesses hydrophobic and oleophobic properties, along with high stability and a prolonged biological half-life, rendering it resistant to degradation and capable of bioaccumulation in the food chain.² PFOS can enter the human body through several pathways,³ such as skin contact, drinking water, and inhalation. Recent animal studies have demonstrated that PFOS is highly toxic to various organisms in terms of reproductive toxicity,⁴ immunotoxicity,⁵ neurotoxicity,⁶ developmental toxicity,⁷ hepatotoxicity,⁸ and cardiotoxicity,⁹ among others. The liver, as a common target organ of poisons, is highly sensitive to foreign substances. Epidemiological studies indicate that PFOS exposure can lead to elevated levels of alanine aminotransferase.¹⁰ While some studies have explored the mechanisms behind PFOS-induced liver damage, the specific mechanism has not been fully elucidated.

The liver is a metabolic organ that plays a crucial role in maintaining metabolic homeostasis.¹¹ Studies have indicated that the onset and progression of certain liver diseases, such as nonalcoholic fatty liver disease, are closely linked to disruptions in liver cholesterol homeostasis.¹² Excessive free cholesterol in the liver can form crystals in liver macrophages, which may trigger oxidative stress, inflammatory response, and

cell death. Hepatic cholesterol homeostasis involves source synthesis, absorption, efflux, and conversion.¹³ Recent research has demonstrated that PFOS disrupts the conversion of cholesterol to bile acids, leading to excessive cholesterol accumulation in the liver.¹⁴ Additionally, PFOS has been shown to increase endogenous cholesterol synthesis.¹⁵ The above studies have shown that PFOS contributes to the dysregulation of liver cholesterol homeostasis, leading to pathological conditions.

Previous studies have indicated that nutritional interventions can effectively reduce the toxicity of environmental pollutants. Flaxseed oil (FO) is an edible oil rich in omega-3 (n-3) polyunsaturated fatty acids, mainly linolenic acid. It offers a range of biological functions, including positive effects on cardiovascular health,¹⁶ improvement of memory impairment,¹⁷ and anti-inflammatory and antioxidant effects.¹⁸ Wang et al.¹⁹ showed that dietary FO can prevent hepatic steatosis induced by alcoholic fatty liver disease by regulating liver lipid homeostasis. Additionally, Han et al.²⁰ reported that FO may improve atherosclerosis in mice by regulating lipid metabolism. However, the effect of FO on PFOS-induced liver

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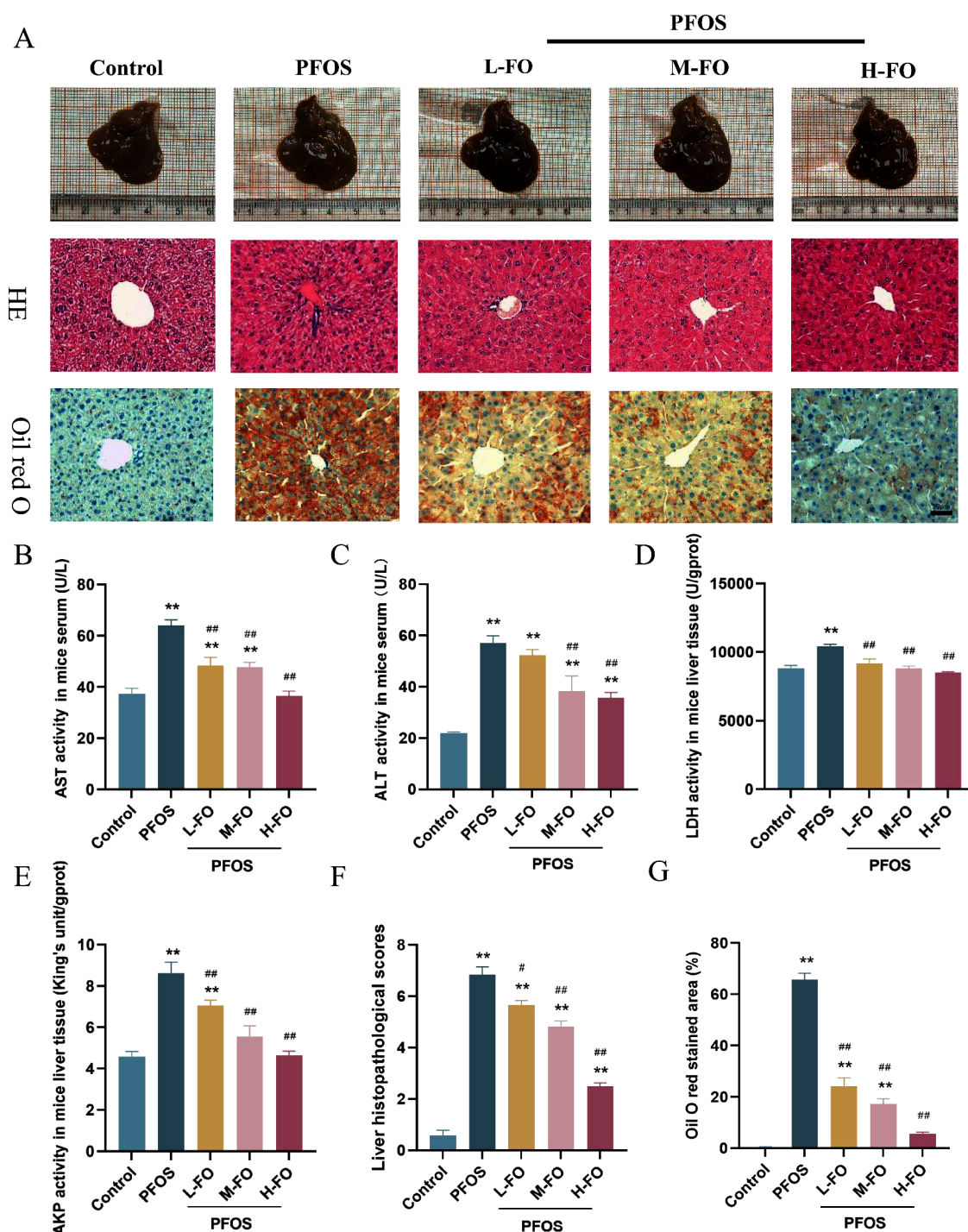


Figure 1. Flaxseed oil reduced liver injury induced by PFOS in mice. (A) Macroscopic pictures of mouse liver in each group, hematoxylin–eosin (HE) staining, and oil red O staining of liver tissue in each group (scale bar = 50 μ m). (B, C) Serum AST and ALT levels in each group. (D, E) Levels of LDH and AKP in livers of the mice. (F) Histopathological score of HE staining. (G) Quantitative analysis of the percentage of oil red O. The values are presented as the mean $\pm$ SEM ($n = 6$). * $p < 0.05$ difference from the control group; ** $p < 0.01$ difference from the control group; # $p < 0.05$ difference from the PFOS group; and ## $p < 0.01$ difference from the PFOS group.

injury remains unclear. This study aims to explore the biological effects and mechanisms of FO in relation to liver injury caused by PFOS through proteomic analysis, with multiple experiments conducted for validation.

MATERIALS AND METHODS

Materials and Reagents. FO and soybean oil were obtained from Aladdin Biochemical Technology Co., Ltd. (Shanghai, China).

PFOS was sourced from Bolingwei Technology Co., Ltd. (Beijing, China). Assay kits for alkaline phosphatase (AKP), lactate dehydrogenase (LDH), alanine aminotransferase (ALT), aspartate aminotransferase (AST), superoxide dismutase (SOD), malondialdehyde (MDA), glutathione (GSH), total antioxidant capacity (T-AOC), total cholesterol (TC), triglyceride (TG), high-density lipoprotein cholesterol (HDL-C), and low-density lipoprotein cholesterol (LDL-C) assay kits were purchased from Nanjing Jiancheng Bioengineering Institute (Nanjing, China). Additionally,

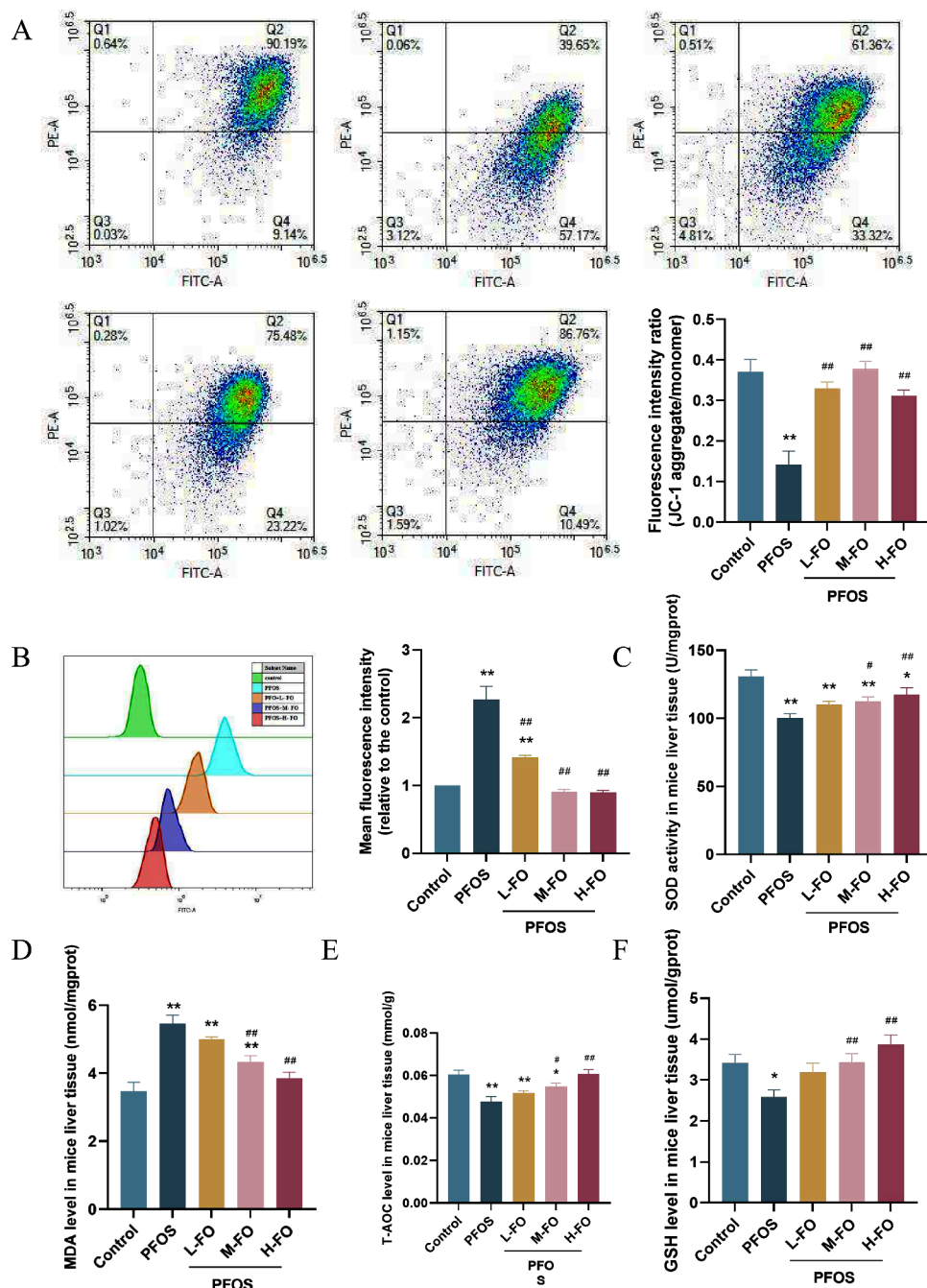


Figure 2. Effects of flaxseed oil on oxidative stress and apoptosis caused by PFOS in liver tissue. Flow cytometric analysis and quantification of mitochondrial membrane potential (A) and ROS (B) in liver of mice. (C–F) Effects of flaxseed oil on SOD, MDA, T-AOC, and GSH levels in the liver. The values are presented as the mean $\pm$ SEM ($n = 6$). * $p < 0.05$ difference from the control group; ** $p < 0.01$ difference from the control group; # $p < 0.05$ difference from the PFOS group; ## $p < 0.01$ difference from the PFOS group.

serum TC, TG, HDL-C, and LDL-C kits were acquired from Shenzhen Mindray Biomedical Electronics Co., Ltd. ELISA kits for interleukin-6 (IL-6), IL-1 β , and tumor necrosis factor- α (TNF- α) were sourced from Shanghai Enzyme-Linked Biotechnology Co., Ltd. (Shanghai, China). Primary antibodies against SREBP2 (#A13049), HMGCR (#A19063), LDLR (#A20808), CYP7A1 (#A10615), and β -actin (#A19056) were purchased from ABclonal (Wuhan, China), while those against CD68 (GB113109-100) were obtained from Servicebio Co., Ltd. (Wuhan, China). Secondary antibodies labeled with horseradish peroxidase were purchased from Abways Technology (Shanghai, China). Details regarding the primary antibody antibodies are provided in Table S1.

Animal Treatment. Thirty C57BL/6J male mice, aged 7 weeks and weighing 22 ± 2 g, were purchased from Beijing Vital River Laboratory Animal Technology. The mice were housed in a pathogen-free facility maintained at a constant temperature (20 to 26 °C), humidity (40 to 60%), and light-controlled conditions (12 h light–dark cycle). They had unrestricted access to water and food in their cages. After 1-week adaptation period, the mice were randomly divided into five groups: (i) control group, (ii) PFOS group, (iii) low-dose flaxseed oil group (L-FO), (iv) medium-dose flaxseed oil group (M-FO), and (v) high-dose flaxseed oil group (H-FO). The control and PFOS groups were administrated with soybean oil daily (2 g/kg). The L-FO group received a mixture of FO and soybean oil in a 1:3 ratio, while the M-FO group received a 1:1 mixture. The H-FO group

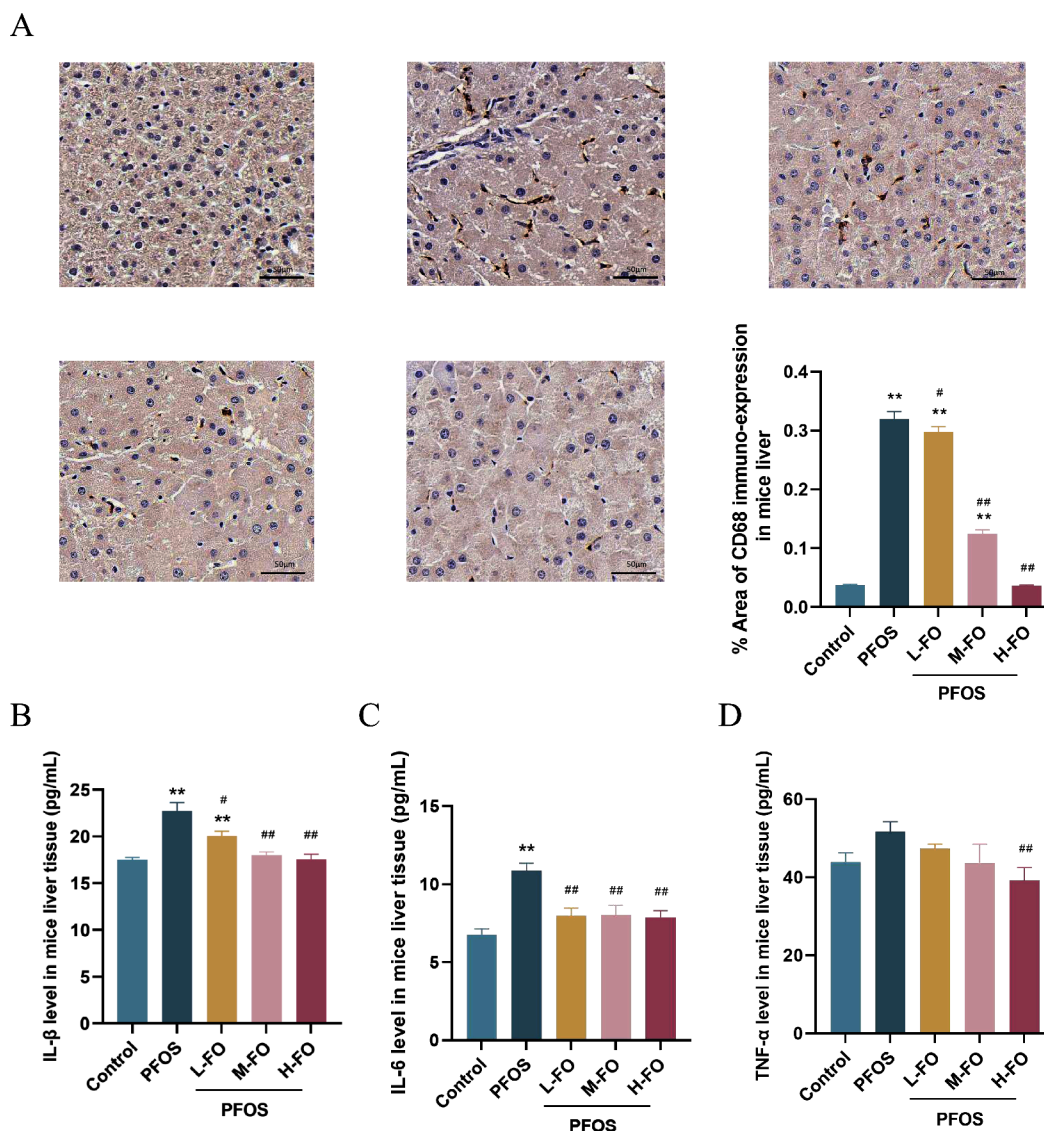


Figure 3. Flaxseed oil reduced positive infiltration of the CD68 macrophage and inflammatory cytokine release caused by PFOS in liver of mice. (A) Representative images of immunohistochemical staining for CD68 and semiquantitative analysis. (B–D) Effects of flaxseed oil on the levels of TNF- α , IL-6, and IL-1 β in liver tissue of mice. The values are presented as the mean $\pm$ SEM ($n = 6$). * $p < 0.05$ difference from the control group; ** $p < 0.01$ difference from the control group; # $p < 0.05$ difference from the PFOS group; and ## $p < 0.01$ difference from the PFOS group.

was given pure FO. All mice were treated via oral gavage for four consecutive weeks (2 g/kg). In addition, the FO intervention and PFOS groups received PFOS daily (10 mg/kg). On the 29th day, all mice were sacrificed, and blood samples and liver tissues were collected for further analyses.

Histological Analysis. A portion of the liver tissue was fixed in 4% paraformaldehyde for paraffin embedding and subsequent hematoxylin–eosin (HE) staining. The extent of tissue inflammation damage was evaluated and scored using a previously established scoring system.²¹ Liver fat accumulation was evaluated using Oil Red O staining of frozen liver tissue.

For immunohistochemistry, paraffin wax-embedded sections (5 μ m) were dewaxed and processed for standard immunohistochemical staining. Briefly, the sections were treated with citrate buffer and blocked with 10% goat serum. They were then incubated overnight with the primary antibody (anti-CD68), followed by incubation with a horseradish peroxidase-conjugated secondary antibody. Images were captured using an Olympus microscope (Tokyo, Japan) and quantified using ImageJ software.

Analysis of Hepatic Lipid and Serum Biochemical Tests. The levels of TG, TC, HDL-C, and LDL-C in the liver, as well as serum

levels of ALT and AST, were measured using assay kits. Serum TC, TG, HDL-C, and LDL-C levels were analyzed using an automatic biochemical analyzer (Mindray BS-830, Shenzhen, China).

Determination of Oxidative Stress and Inflammation in Liver Tissue. Multiple markers of oxidative stress, including SOD, GSH, T-AOC, and MDA, were detected in liver tissue using a test kit. Enzyme-linked immunosorbent assay (ELISA) was employed to quantify the levels of IL-1 β , IL-6, and TNF- α in mouse liver tissue. All experimental conditions were conducted in strict accordance with the instructions.

Proteomics Analysis. Liver tissue was immediately digested and quantitatively analyzed using data-independent acquisition proteomics technology (three groups of four mice each). Analyses were performed using LC–MS with UPLC M-Class systems (Waters, USA) and a Q Exactive HF mass spectrometer (Thermo Fisher, USA). A P value < 0.05 and a fold change (FC) > 1.2 were considered significant, with < 0.83 indicating downregulation and > 1.2 upregulation. Volcano plots, heat maps, GO analyses, KEGG analyses, and relevant figures were generated using online analysis software (<https://www.omicsolution.org/wkomics/main/>). The protein inter-

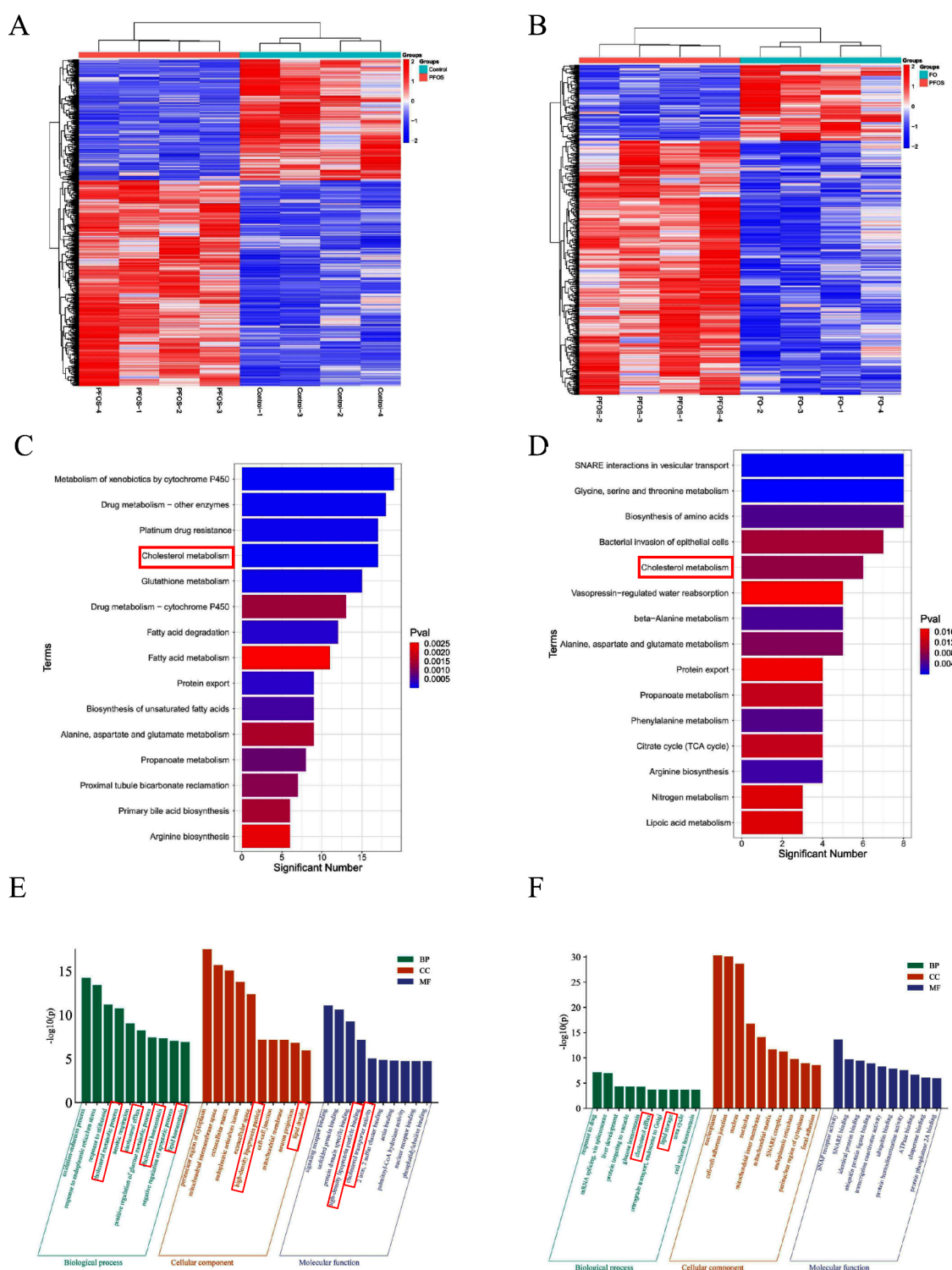


Figure 4. Proteomic analysis of mouse liver. (A) Heat map of differential proteins between control and PFOS groups. (B) Heat map of differential proteins between PFOS group and flaxseed oil group. (C) Bar graph of KEGG pathway analysis between control and PFOS groups. (D) Bar graph of KEGG pathway analysis between PFOS group and flaxseed oil group. (E) GO function annotation analysis between control and PFOS groups. (F) GO function annotation analysis between PFOS group and flaxseed oil group.

action network was created using the STRING online Web site (<https://string-db.org/>).

Western Blot Analysis. Mouse liver tissues were lysed using RIPA buffer containing protease inhibitors, and the protein supernatants were obtained by centrifugation at 12,000 rpm for 10 min at 4 °C. The supernatant was then denatured by boiling in a metal bath with 5× buffer loading buffer for 5 min. Protein samples

were separated using 8% sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS–PAGE) and electro transferred to PVDF membranes. The membranes were blocked with 5% skim milk for 2 h and then cut into strips and incubated with primary antibodies overnight at 4 °C. The following day, the bands were washed three times with TBST and incubated with the corresponding secondary antibodies for 40 min at room temperature. The bands were washed

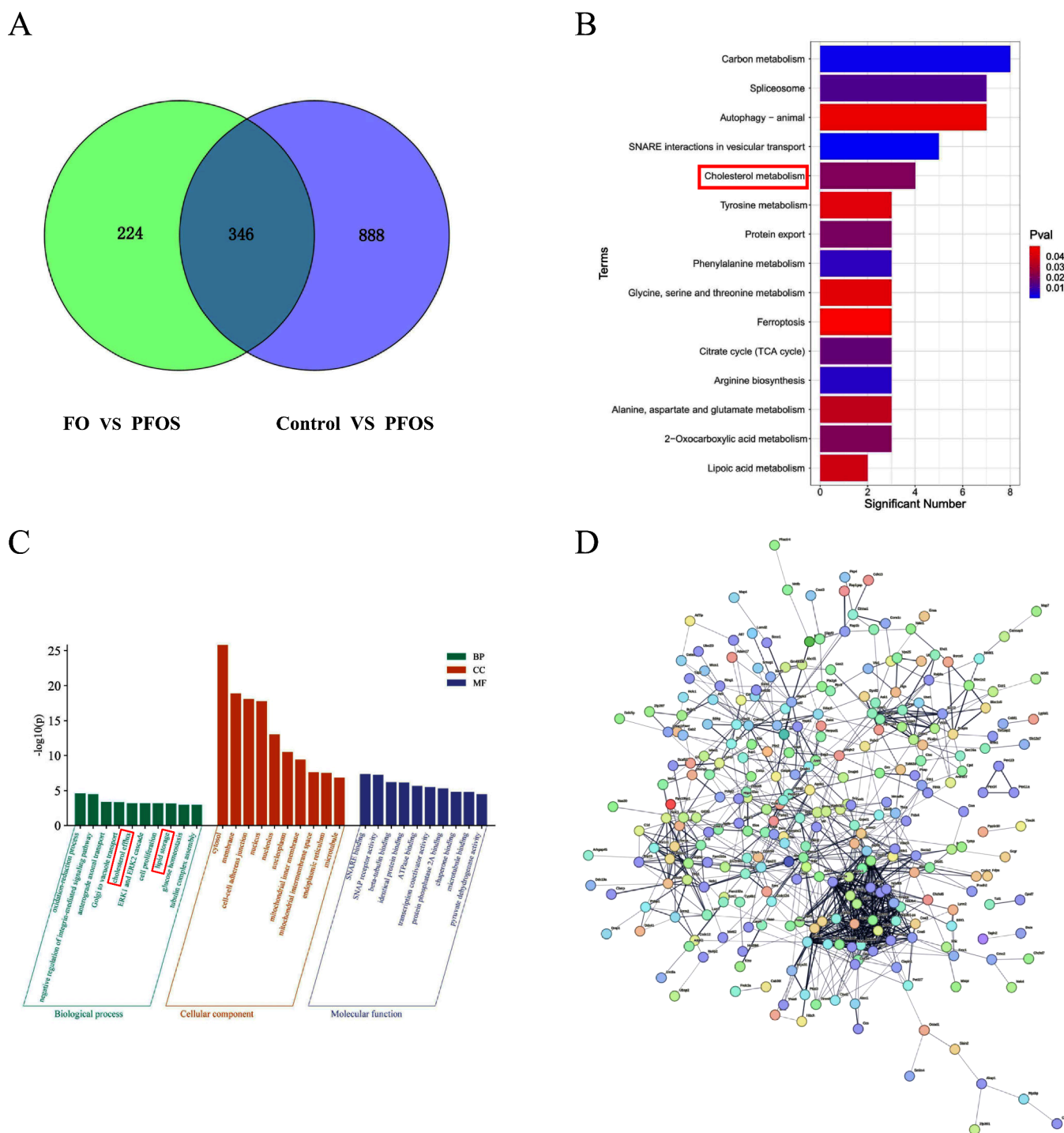


Figure 5. DEPs in the control and PFOS groups intersected with those in the PFOS and flaxseed oil groups. (A) Venn diagram of intersection analysis. (B) Bar graph of KEGG pathway analysis of the intersection DEPs. (C) GO functional annotation of the intersection DEPs. (D) Protein–protein interaction network analysis.

three additional times with TBST and detected using an ECL detection reagent. Finally, ImageJ software was used to quantify the density of the bands, and expression levels were normalized using β -actin as an internal control.

Statistical Analysis. Statistical analysis was conducted using SPSS 21.0 software and GraphPad Prism software. All numerical data are represented as mean $\pm$ standard error of the mean (SEM), and one-way ANOVA was used to analyze the mean differences between groups. P values <0.05 were considered statistically significant.

RESULTS

FO Alleviated PFOS-Induced Hepatic Injury in Mice.

First, the pathological changes in liver tissue were observed to investigate the role of FO in PFOS-induced liver injury. The macroscopic examination revealed that FO could reduce liver enlargement caused by PFOS (Figure 1A). Additionally, pathomorphological observations were performed under HE and red O staining of mouse liver. The results showed that PFOS exposure induced tissue edema, hemorrhage, inflamma-

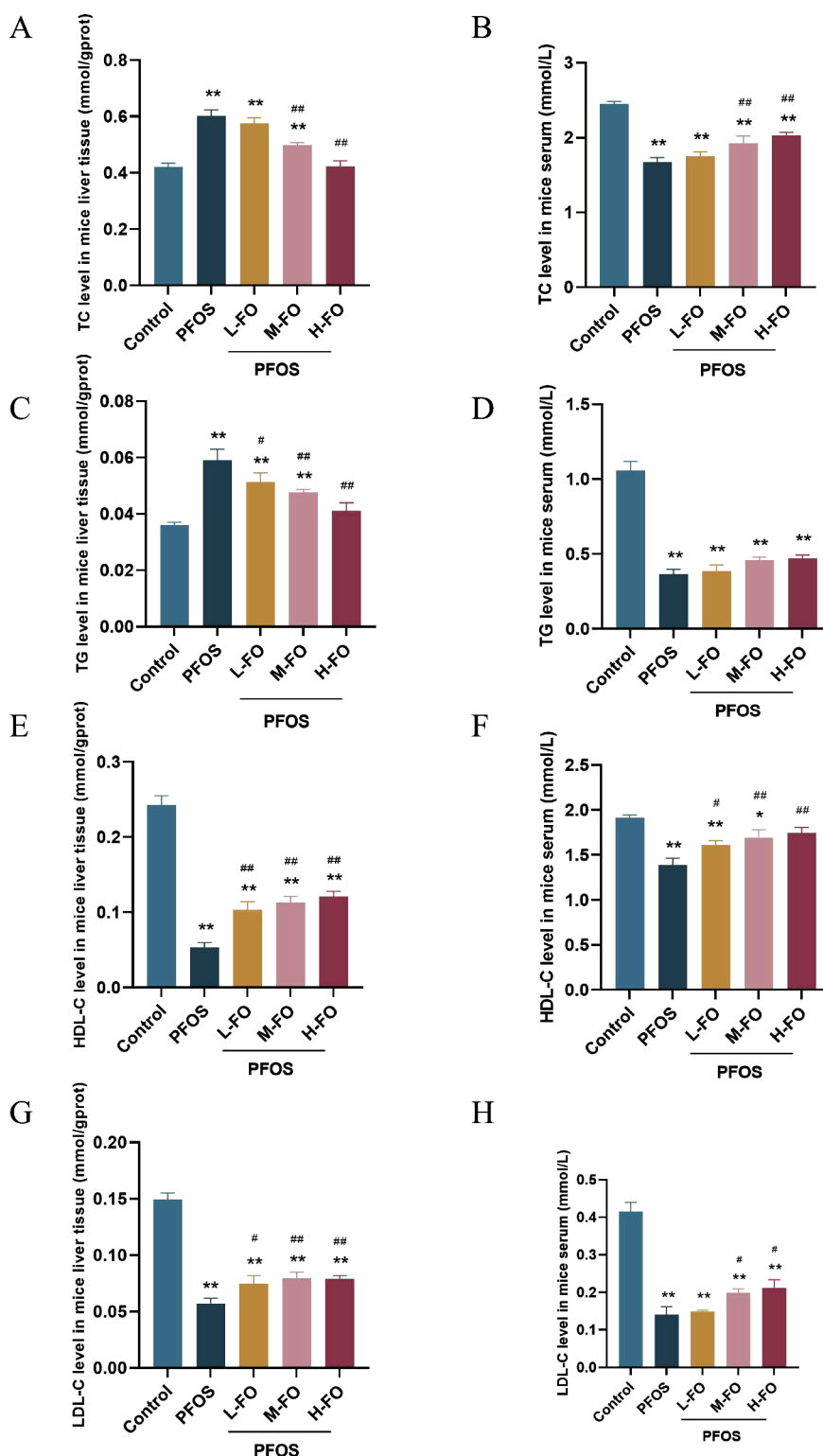


Figure 6. Flaxseed oil modulated serum and hepatic lipid changes induced by PFOS in liver and serum samples of the mice. Determination of TC (A, B), TG (C, D), HDL-C (E, F), and LDL-C (G, H) levels in the liver and serum. The values are presented as the mean $\pm$ SEM ($n = 6$). * $p < 0.05$ difference from the control group; ** $p < 0.01$ difference from the control group; # $p < 0.05$ difference from the PFOS group; and ## $p < 0.01$ difference from the PFOS group.

tory cell infiltration, and lipid accumulation. FO supplementation alleviated histopathological alterations of mice treated with PFOS (Figure 1A), as semiquantitatively demonstrated by histopathological scores and the ratio of lipid droplet area (Figure 1F,G). Furthermore, serum and liver markers

indicative of liver damage, including AST, ALT, LDH, and AKP levels, were measured (Figure 1B–E). The results indicated that, compared to the control group, the levels of AST, ALT, LDH, and AKP were significantly elevated in the PFOS exposure group. However, FO at varying doses reversed

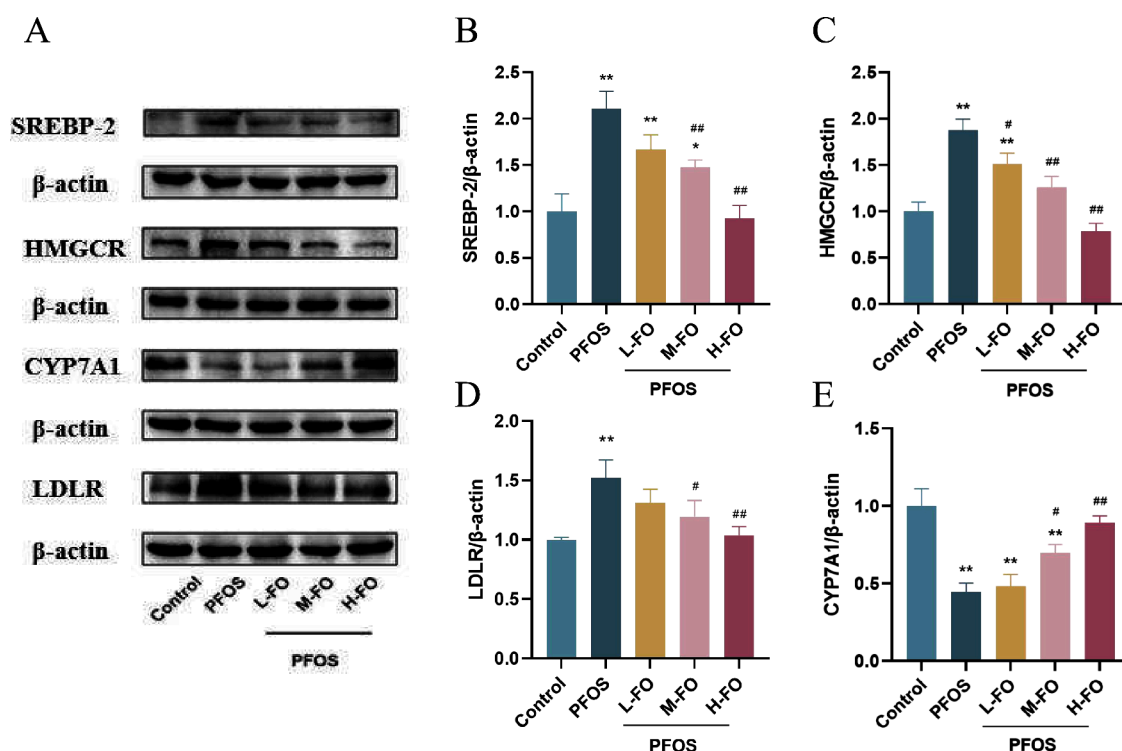


Figure 7. Flaxseed oil ameliorated PFOS-induced disorder of cholesterol metabolism in liver of mice. (A–E) Western blot results and quantitative analysis of proteins involved in cholesterol metabolism, including SREBP2, HMGCR, LDLR, and CYP7A1. The values are presented as the mean $\pm$ SEM ($n = 6$). * $p < 0.05$ difference from the control group; ** $p < 0.01$ difference from the control group; # $p < 0.05$ difference from the PFOS group; and ## $p < 0.01$ difference from the PFOS group.

these changes (Figures 1B–E). Overall, these data suggest that FO supplementation effectively attenuates liver damage caused by PFOS exposure.

FO Attenuated Apoptosis and Oxidative Stress in Mice. Mitochondrial membrane potential (MMP) and various indices of oxidative stress were used to further assess the effect of FO on hepatocyte apoptosis and oxidative stress. Results revealed that the proportion of apoptotic cells increased in the PFOS group compared to the control group, which was reversed following FO supplementation (Figure 2A). Elevated ROS levels are crucial indicators of oxidative stress. The results indicated that ROS accumulation occurred in hepatocytes after PFOS exposure, with ROS levels in the FO group significantly lower than those in the PFOS group. Moreover, several antioxidant enzymes (SOD, T-AOC, and GSH) and MDA were measured. As shown in Figure 2C–F, PFOS administration resulted in increased MDA levels and decreased SOD, T-AOC, and GSH levels. Notably, FO supplementation reversed these effects. Therefore, FO exerts antiapoptotic and antioxidative effects in response to PFOS exposure.

FO Suppressed the Infiltration of Liver CD68-Positive Macrophages and Inflammatory Response Induced by PFOS Exposure. The overactivation of macrophages is crucial in liver diseases. CD68 (macrosialin) is a macrophage-specific marker and biomarker of macrophage infiltration. This study performed IHC staining to examine the distribution of macrophages in liver tissue. As shown in Figure 3A, the positive area of CD68 staining was higher in the PFOS exposure group than that of the control group, revealing a reduction in FO supplementation. This result indicated that pretreatment with FO reduced the activation of hepatic macrophages induced by PFOS exposure. Moreover, the

expressions of inflammatory factors in the liver of mice, including IL-6, IL-1 β , and TNF- α , were detected. Compared with the control group, PFOS exposure increased the contents of IL-6, IL-1 β , and TNF- α in the liver of mice, and FO supplementation attenuated the contents of these inflammatory factors in the liver (Figure 3B–D). These results indicated that FO reduced PFOS-induced liver damage by mitigating macrophage activation.

Proteomic Analysis. Proteomic analysis was conducted to further elucidate the potential liver-protective mechanisms of FO. Pairwise Pearson correlation coefficients for 12 samples (four replicate $\times$ 3 groups) were calculated to assess the relative quantitative reproducibility of liver proteins in each group. The results, shown in Figure S1C, demonstrate strong within-group correlations ($r > 0.80$). As illustrated in Figure S1B, 1234 differentially expressed proteins (DEPs) (777 up-regulated and 457 down-regulated) between the control and PFOS groups were identified. Additionally, 570 DEPs (439 up-regulated and 131 down-regulated) were identified in the FO group compared with PFOS group. The heatmap based on these DEPs also showed clear differences between the PFOS and control groups (Figure 4A). Similar results were observed between the PFOS and FO groups (Figure 4B). KEGG enrichment analysis was performed to further identify the biological function of key targets. The KEGG enrichment analysis of DEPs between the control and PFOS groups indicated that these DEPs were mainly enriched in amino acid and cholesterol metabolisms (Figure 4C). Interestingly, KEGG enrichment analysis of DEPs between the FO and PFOS groups also highlighted in cholesterol metabolism (Figure 4D and Tables S2 and S3). The GO functional annotation of the selected DEPs enriched the biological importance of their

DEPs. Ten terms with significant differences in BP, CC, and MF (Figure 4E,F and Tables S5 and S6) were listed. Notably, many GO terms related to metabolic processes were identified. GO analysis of DEPs between the control and PFOS groups revealed terms such as cholesterol metabolic process, cholesterol efflux, cholesterol homeostasis, and lipid homeostasis in BP. In CC, terms such as high-density lipoprotein particle and lipid droplet are associated with cholesterol synthesis and transport. In MF, high-density lipoprotein particle binding and cholesterol transporter activity were identified (Figure 4E). In the GO analysis of PFOS and FO groups, cholesterol efflux and lipid storage were also identified in BP, indicating their relation to cholesterol and lipid disorders (Figure 4F).

Data analysis was performed on the intersection of two different protein comparisons to further determine the protective mechanism of FO against PFOS-induced liver injury. A total of 346 DEPs were identified to perform KEGG and GO analyses (Figure 5A). KEGG enrichment analysis results showed that DEPs were also mainly enriched in cholesterol metabolism (Figure 5B and Table S4). GO analyses demonstrated the same results. The identification of biological processes was associated with cholesterol efflux and lipid storage (Figure 5C and Table S7). Additionally, the interaction network was further performed, and the core value was calculated (Figure 5D). The above results indicated that FO may attenuate PFOS-induced liver injury by regulating cholesterol metabolism.

FO Modulated Serum and Hepatic Lipid Changes Induced by PFOS Exposure. The levels of TC, TG, HDL-C, and LDL-C in serum and liver of mice were measured to further determine the effect of FO on the changes of lipid levels in mice induced by PFOS. These results indicated that PFOS administration increased the levels of TC and TG in the liver of mice, while FO administration reduced these levels (Figure 6A,C). In serum, TC and TG levels were found to be lower in the PFOS group, but FO supplementation elevated these levels (Figure 6B,D, respectively). Regarding HDL-C and LDL-C levels, both were significantly decreased after PFOS intervention in the serum and liver of mice, but this decrease was ameliorated by FO supplementation (Figures 6E–H). Overall, the abovementioned data demonstrate that FO supplementation can improve lipid disorders caused by PFOS.

FO Ameliorated PFOS-Induced Disorder of Cholesterol Metabolism. The results of the proteomic analysis indicated that PFOS exposure disrupted hepatic cholesterol metabolism. Proteins related to cholesterol metabolism were evaluated to further validate the effect of FO on cholesterol metabolism induced by PFOS exposure. SREBP2 is a key regulator of cholesterol synthesis, and inhibiting its activity can reduce cholesterol synthesis. SREBP2 regulates cholesterol synthesis in the liver by modulating the expression of LDLR and HMGCR proteins. PFOS exposure increased the expression of SREBP2, LDLR, and HMGCR, while FO supplementation inhibited the expression of the three proteins (Figure 7A–D). CYP7A1, a rate-limiting enzyme for bile acid synthesis, is crucial for maintaining bile acid homeostasis. The results showed that PFOS decreased CYP7A1 expression, but FO administration increased its level in the liver (Figure 7E). Therefore, these results indicate that FO can ameliorate the disruption of cholesterol metabolism induced by PFOS exposure.

DISCUSSION

Nutritional intervention is considered one of the most effective strategies to prevent diseases related to environmental toxins. This study found that FO could attenuate hepatotoxicity induced by PFOS exposure, as evidenced by reduced inflammatory cell infiltration, inflammatory cytokine production, and lipid droplet accumulation. A proteomic analysis was conducted to further explore the protective mechanism of FO on PFOS-induced liver injury. The results indicated that the protective mechanism of FO may be related to cholesterol metabolism. Moreover, PFOS exposure decreased the expression of CYP7A1 while increasing the expression of SREBP2, HMGCR, and LDLR; however, FO supplementation significantly reversed the effect.

In the past decade, the health risks of PFOS to humans have attracted increasing attention.²² Current studies have shown that PFOS can persist in the environment and accumulate in human serum, breast milk, kidneys, and liver through ingestion or exposure to products containing PFOS. According to the European Food Safety Authority report (2008), the tolerable daily intake (TDI) of PFOS is 150 ng/kg bw/day.²³ However, this value is difficult to achieve for occupationally exposed populations. One epidemiological study indicated that the serum PFOS level in fishery workers and production workers can reach 31,400 ng/mL.²⁴ A study found that the serum PFOS level in C57BL/6 mice at a dose of 7 mg/kg/day for 28 days was approximately 8434 ng/mL.²⁵ Therefore, in the current study, the dosage of PFOS administered to animal models was similar to the occupational exposure level in humans. As previously stated, evidence indicating that the liver is the primary target organ for the toxic effects induced by PFOS is growing.^{26–29} Studies have shown that PFOS is rarely metabolized by the liver or other metabolic organs because it is chemically stable and does not undergo any chemical reactions.^{30,31} Marques et al.³² showed that PFOS can cause liver steatosis in humans, and a study on mice also revealed that PFOS mainly accumulates in the liver, followed by the lung, kidney, spleen, heart, and brain.³³ The current results also showed that infiltration of the inflammatory cells, lipid deposition, and oxidative stress increased in the liver of PFOS-exposed mice, as evidenced by the histopathologic staining and the kit test results.

Understanding the mechanism of PFOS-induced liver disease progression is crucial for developing effective therapeutic strategies against liver injury caused by PFOS. A review by Wang et al.³⁴ showed that the impact of PFOS on the liver is related to multiple signaling pathways, including inflammation, oxidative stress, PPAR-related signaling pathway, apoptosis, and autophagy. FO is a common dietary oil that is rich in ALA and has antioxidant and anti-inflammatory effects.^{35,36} Previous studies have shown that PFOS exposure can increase the expression of TNF- α and IL-6 by activating the NF- κ B signaling pathway.³⁷ The obtained results are consistent with previous studies. The results showed that PFOS exposure increased the levels of inflammatory factors (TNF- α , IL-6, and IL-1 β) in liver tissue. However, the levels of inflammatory factors were significantly decreased after treatment with FO. ROS are important pathogenic factors in the pathogenesis of numerous liver diseases, and persistent excessive ROS can be toxic to cells.³⁸ The findings from the previous study showed that PFOS exposure increased ROS production in the liver and disrupted the balance between

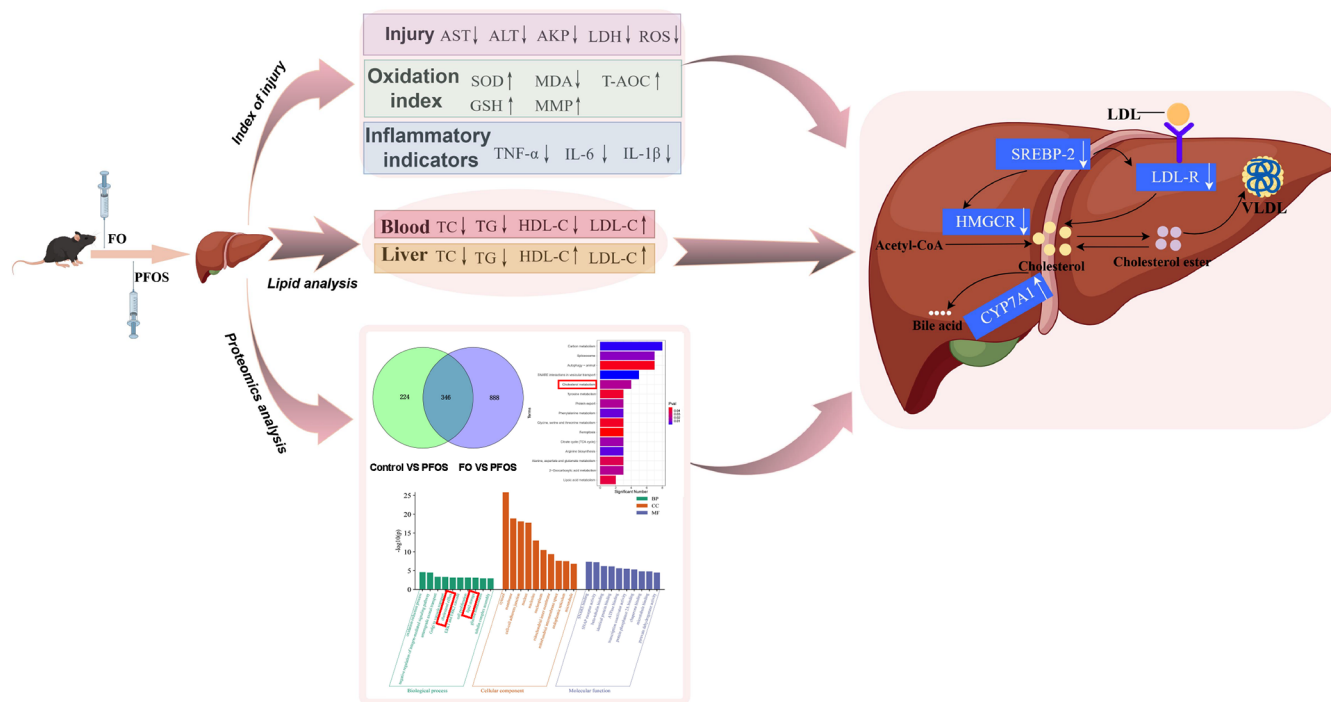


Figure 8. Flaxseed oil supplementation attenuates PFOS-induced liver injury mechanisms. PFOS mediates liver injury by disrupting hepatic cholesterol metabolism, which can be counteracted by flaxseed oil.

oxidative and antioxidant systems.² This study revealed that PFOS exposure led to a reduction in the levels of GSH, SOD, and T-AOC, while also increasing the ROS accumulation. These effects were mitigated by treatment with FO. Studies have shown that excessive ROS can inhibit MMP and trigger apoptosis.³⁹ This study also examined MMP, and the results indicated that FO inhibited PFOS-induced apoptosis.

Investigation of proteomics can provide a material basis for understanding the mechanisms of life activities and offer new insights into the pathogenesis of various diseases. Differential analysis of the identified proteins in the liver was performed, followed by KEGG enrichment and GO analyses on the DEPs between the PFOS and the control, FO, and PFOS groups and their intersections. The KEGG enrichment analysis results showed that DEPs were enriched in cholesterol metabolism pathways. Interestingly, the GO analysis identified numerous GO terms related to cholesterol metabolic, including cholesterol metabolic process, cholesterol efflux, cholesterol homeostasis, and lipid homeostasis. The above proteomics results indicated that FO may alleviate PFOS-induced liver injury by regulating hepatic cholesterol metabolism. The liver is the major site of cholesterol synthesis, accounting for approximately 50% of the synthesis.⁴⁰ Cholesterol synthesis is regulated by SREBP2 and HMGCR. SREBP2 binds to a sterol regulatory element in the promoter of its target gene, activating HMGCR transcription.⁴¹ Bao et al.⁴² found that the linseed peptide could reduce endogenous cholesterol synthesis in HepG2 by reducing the expressions of SREBP2 and HMGCR. However, whether FO can alleviate PFOS-induced liver injury by regulating cholesterol metabolism remains unclear. This study found that FO notably increased the HDL-C levels in liver and serum of mice compared with the PFOS group. In the detection of cholesterol synthesis-related protein expression, PFOS exposure increased hepatic SREBP2 and HMGCR expressions, which were then decreased in a dose-dependent

manner after FO administration. LDLs comprise very low-density lipoproteins (VLDLs) that deliver endogenous lipids, including triglycerides and cholesterol, from the liver to peripheral tissues.^{43,44} Wang et al. indicated that PFOS exposure can interfere with VLDL synthesis and excretion.⁴⁵ The results of this study reveal that the transport of endogenous lipids from the liver to peripheral tissues is hindered, resulting in elevated TG levels within the liver and reduced TG levels in the serum. These findings are supported by the experimental data obtained in this study. However, FO supplementation attenuated this effect, indicating that FO prevented PFOS from interfering with VLDL synthesis and excretion. SREBP2 also regulates LDLR expression,¹³ which mediates the binding and endocytosis of circulating lipoproteins, especially LDL-C. Furthermore, this study found that PFOS increased LDLR expression while simultaneously inhibiting VLDL synthesis and excretion in the liver, thereby reducing LDL levels. This reduction indicates that the binding and endocytosis of LDL-C mediated by the enhanced LDLR expression could not offset the reduction of LDL levels caused by PFOS. LDLR expression was decreased after FO supplementation, which reduced the endocytic absorption of cholesterol and alleviated the pressure of cholesterol accumulation in the liver. The main pathways of cholesterol catabolism in the liver involve the synthesis and excretion of bile acids. CYP7A1 is the rate-limiting enzyme in classic pathways for hepatic bile acid synthesis. Previous studies have shown that PFOS can impair bile acid metabolism and reduce CYP7A1 protein expression in human HepaRG hepatoma cells.¹⁴ The present study also demonstrated that CYP7A1 expression was decreased in PFOS-exposed mice and FO supplementation significantly increased CYP7A1 expression. Thus, FO can increase the conversion of cholesterol into bile acids, thereby mitigating the buildup of excess cholesterol in the liver.

The authors believe that this study was the first to elucidate FO for alleviating PFOS-induced liver injury by regulating cholesterol metabolism. The selected dose of FO was determined in accordance with the recommended intake for humans, which is of considerable importance to guide the diet of residents. FO is an abundant source of ALA. In this study, when the doses of ALA administered to mice were converted to human doses (assuming the mice weighed 20 g and using a conversion factor of 0.081 between mice and humans), the applied doses for a 60 kg human were 1.215, 2.43, and 4.86 g day⁻¹. These doses are similar to the recommended dosages⁴⁶ for modest dietary consumption of ALA, which have been suggested for the primary and secondary prevention of diseases.^{47,48} However, this study has some limitations. First, the method of establishing a model via oral gavage of PFOS cannot completely replace the real-world human PFOS exposure. Second, cholesterol homeostasis is an extremely complex process, and the specific regulatory mechanism of FO on cholesterol metabolism still needs further research. Lastly, the regulation of cholesterol metabolism may be a possible mechanism through which FO exerts its hepatoprotective effect. Other undisclosed molecular mechanisms warrant further study.

Overall, this study demonstrated that FO supplementation had a beneficial effect on PFOS-induced hepatotoxicity, and the mechanism might be related to the regulation of hepatic cholesterol metabolism (Figure 8). Therefore, FO supplementation can be used as an effective nutritional intervention strategy against hepatic damage induced by PFOS. These findings emphasize the importance of further research on the impact of nutritional intervention strategies in mitigating the negative health consequences of environmental exposure.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jafc.4c04438>.

(Figure S1) Distribution of DEPs; (Table S1) primary antibodies used in Western blot; (Table S2) pathways enriched in KEGG analysis for DEPs between control and PFOS groups; (Table S3) pathways enriched in KEGG analysis for DEPs between PFOS and FO groups; (Table S4) KEGG enrichment analysis was used to analyze the pathways of intersection DEPs; (Table S5) major terms enriched in GO analysis of DEPs in the control and PFOS groups; (Table S6) major terms enriched in GO analysis of DEPs in the PFOS and FO groups; (Table S7) major terms enriched in GO analysis of intersection DEPs (PDF)

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